Higher-Degree Polynomial Families on $x^4 + y^3 + z^2 + 1 = 0$: Covariant Syzygies, Resolution of Open Degree-$(3, 8, 12)$ and $(3, 10, 15)$ Systems, and Universal Bounds for $\deg x \geq 3$

Research Continuation Report

Extending the results of *Integer solutions of* $x^4 + y^3 + z^2 + 1 = 0$ (15 October 2026)

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Abstract

We extend the study of polynomial solutions $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to the affine surface $S : x^4+y^3+z^2+1 = 0$ (written $Y^3-z^2 = x^4+1$ with $Y = -y$) into the previously open regime $\deg x \geq 3$ for arbitrary y and z. In prior work, polynomial families were excluded for $\deg x \leq 2$ (Theorem A) and for $\deg x = 3, \deg y \leq 4$ (Theorem B), while the cases $\deg x = 3, \deg y \in \{8, 10, 14, 16, 20, 22\}$ and $\deg x \geq 4$ remained open; in particular, the degree-$(3, 8, 12)$ polynomial system of 11 equations in 10 unknowns did not terminate under standard Gröbner basis routines, and boundary dimension counts led to the heuristic expectation that no polynomial curves exist for $\nu = \frac{1}{2}\deg y \geq 4$ even over $\overline{\mathbb{Q}}$.

We establish six new results: (1) for *any* $\deg x = d_x \geq 1$ and arbitrary $y(t), z(t) \in \mathbb{R}[t]$, positivity of leading terms forces $\deg y \geq \lceil 4d_x/3 \rceil$, while the triviality of the Mordell–Weil group $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ forces the Galois trace zero-cycle $\sum_{i=1}^{d_x}[P_i]$ to vanish on the generic fibre, yielding a canonical divisorial reduction of $z(t)$ modulo $\mathbb{Q}[x, Y]$; (2) for arbitrary $d_x = \deg x \geq 3$, the Davenport–Stothers–Zannier and Mason–Stothers inequalities confine all nonconstant solutions to either the canonical regime $(d_x, d_Y, d_z) = (3k, 4k, 6k)$ or the Hall–Davenport regime $d_Y = 2\nu, d_z = 3\nu$ with the sharp universal upper bound $d_Y \leq 8d_x - 2$ ($\nu \leq 4d_x - 1$); (3) for $\deg x = 3$ and arbitrary $\deg y = 2\nu$, we prove that the trace-zero characteristic polynomial coefficients $p(u)$ and $N(u)$ are the Hessian and Jacobian covariants of the binary cubic index form $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$, satisfying the syzygy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$, and that the unified identity $\Phi(u) \equiv 0$ admits a bihomogeneous $\mathbb{G}_{\mathrm{m}}^2$-action that triangularizes the top six equations linearly; (4) using this covariant triangularization, we completely solve the previously intractable open system for $\nu = 4$ ($\deg x = 3, \deg y = 8, \deg z = 12$), proving that no solution exists over $\mathbb{Q}$, while discovering an explicit non-trivial solution over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ that disproves the heuristic emptiness expectation over $\overline{\mathbb{Q}}$; (5) we prove the nonexistence of solutions in the constant-m branch for $\nu = 5$ ($\deg x = 3, \deg y = 10, \deg z = 15$) via an exact resultant reducing to an irreducible quintic; and (6) we show that our new $(3, 8, 12)$ identity arises from a degree-$(2, 3)$ section of the pullback rational elliptic surface $\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$ of Mordell–Weil lattice D_4^*, generating infinite towers of polynomial identities over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ for all $k \geq 1$.

1 Introduction and Context

Let $S \subset \mathbb{A}^3_{\mathbb{Q}}$ denote the smooth affine surface defined by

$$x^4 + y^3 + z^2 + 1 = 0, \qquad \text{or equivalently} \qquad Y^3 - z^2 = x^4 + 1 \quad (Y = -y). \tag{1}$$

Whether $S(\mathbb{Z})$ is infinite is a well-known open problem highlighted by Grechuk [1, 2]. In the companion paper [3], viewing S as the affine complement of the zero section O and the type-IV fibre F_∞ on the rational elliptic surface $\pi : \hat{E} \to \mathbb{P}^1_x$ with Mordell–Weil lattice $\mathrm{MW}(\hat{E}/\overline{\mathbb{Q}}(x)) \cong E_6^*$ [4, 5] and Frobenius-trace verification $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = 0$ established two foundational nonexistence theorems for polynomial families $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$:

Result	Tag	Summary and Section Reference
Theorem 2.1 (Lower Bound & Trace)	[E]	For all $d_x = \deg x \geq 1$, $\deg y \geq \lceil 4d_x/3 \rceil$ over $\mathbb{R}$, and $\sum_{i=1}^{d_x}[P_i] = O$ in $E(\mathbb{Q}(x))$ gives a canonical divisorial reduction of $z(t)$ (Section 2).
Theorem 2.3 (Universal Upper Bound)	[E]	For all $d_x \geq 3$ and any $(y,z) \in \mathbb{C}[t]^2$, $\deg y = 2\nu \leq 8d_x - 2$ ($\nu \leq 4d_x - 1$); if $d_x = 3k$, either $\deg y = 4k$ or $2\nu > 4k$ (Section 2).
Theorem 3.1 (Covariant Syzygy)	[E]	$p = \frac{1}{12}\operatorname{Hess}(I)$ and $N = -\frac{1}{9}\operatorname{Jac}(I,p)$ satisfy $4p^3 + 27N^2 = (27u^2 + 4s^3)I^2$; $\Phi(u) \equiv 0$ is $\mathbb{G}_{\mathrm{m}}^2$-bihomogeneous in degrees $u^{\geq 5}$ (Section 3).
Theorem 4.1 (Explicit $(3,8,12)$ Curve)	[E]	Explicit polynomial solution of degrees $(3,8,12)$ over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ for $\nu = 4$, refuting emptiness of $\nu \geq 4$ moduli over $\overline{\mathbb{Q}}$ (Section 4.1).
Theorem 4.2 (Exclusion of $\nu = 4$ over $\mathbb{Q}$)	[E]/[C]	Complete resolution of the open $\deg x = 3, \deg y = 8, \deg z = 12$ system over $\mathbb{Q}$ in both $m_1 = 0$ and $m_1 \neq 0$ branches (Sections 4.1–4.2).
Theorem 5.1 (Exclusion of $\nu = 5, m_1 = 0$)	[E]/[C]	Complete elimination of $\deg x = 3, \deg y = 10, \deg z = 15$ for constant m via overdetermined $\mathbb{G}_{\mathrm{m}}^2$-invariants and an irreducible quintic (Section 5).
Theorem 6.1 (Pullback Towers for $d_x = 3k$)	[E]	Pullback to $\tilde{z}^2 = \tilde{Y}^3 - (w^3+1)$ yields families of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ for all $k \geq 1$ (Section 6).

Table 1: Summary of new theorems and verification status ([E] = complete elementary/analytic proof by hand; [C] = exact symbolic computation in SymPy).

- **Theorem A ($\deg x \leq 2$):** There is no nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ with $\deg x \leq 2$ for any y, z.
- **Theorem B ($\deg x = 3, \deg y \leq 4$):** There is no nonconstant polynomial solution in $\mathbb{Q}[t]^3$ with $\deg x = 3$ and $\deg y \leq 4$ (which forces degrees $(3,4,6)$).

However, Section 7 of [3] left two major regimes open:

1. **The higher-degree regime for $\deg x = 3$ ($\deg y \geq 5$):** Proposition 7.1 of [3] showed that any solution with $\deg x = 3$ and $d = \deg Y \geq 5$ must satisfy $d = 2\nu \in \{8, 10, 14, 16, 20, 22\}$ ($\nu \in \{4, 5, 7, 8, 10, 11\}$). For the very first open case $\nu = 4$ ($d = 8, \deg z = 12$), eliminating $n(x)$ via $4m^2e_3 - e_2^2 - 4m^2(x^4+1) = 0$ produced a system of 11 equations of degree up to 8 in 10 unknowns for which Gröbner basis computations in Singular modulo 32003 failed to terminate, while boundary dimension counts in Appendix C of [3] gave expected dimension $3 - \nu < 0$, motivating the heuristic conjecture that the moduli spaces for $\nu \geq 4$ are empty even over $\overline{\mathbb{Q}}$.
2. **The regime $\deg x \geq 4$ for arbitrary y and z:** Nothing was proved for $\deg x \geq 4$, where the relevant divisor classes have positive expected dimension growing linearly with $\deg x$.

In this paper, we develop a systematic algebraic and geometric framework that extends the theory to $\deg x \geq 3$ for arbitrary $y(t)$ and $z(t)$, completely resolves the open $\nu = 4$ ($\deg y = 8$) system and the constant-m branch of $\nu = 5$ ($\deg y = 10$) for $\deg x = 3$, and reveals a surprising geometric phenomenon over $\overline{\mathbb{Q}}$: the $\nu = 4$ moduli space is *not* empty over $\overline{\mathbb{Q}}$, but carries an explicit polynomial identity over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ arising from a rational elliptic surface pullback that generates infinite towers of polynomial solutions for $\deg x = 3k$.

2 Universal Bounds and Divisorial Reduction for Arbitrary $\deg x \geq 3$

We begin by establishing general structural theorems valid for arbitrary degree $d_x = \deg x \geq 1$ and arbitrary polynomials $y(t), z(t)$.

2.1 Positivity Lower Bound and Trace-Zero Divisorial Decomposition

Theorem 2.1 (Positivity Lower Bound and Trace-Zero Divisorial Reduction)**.** *Let $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ be a nonconstant solution to $x^4 + y^3 + z^2 + 1 = 0$, and put $Y(t) = -y(t)$, $d_x = \deg x \geq 1$, $d_Y = \deg Y$, and $d_z = \deg z$.*

(i) ***Universal lower bound over $\mathbb{R}$ and $\mathbb{Q}$:*** *For every nonconstant solution in $\mathbb{R}[t]^3$ (and hence in $\mathbb{Q}[t]^3$), the leading terms of $x(t)^4$ and $z(t)^2$ cannot cancel, so $3d_Y = \max(4d_x, 2d_z) \geq 4d_x$. Consequently,*

$$\deg y = d_Y \geq \left\lceil \frac{4d_x}{3} \right\rceil, \qquad \deg z = d_z \leq \frac{3}{2} d_Y. \tag{2}$$

In particular, if $3 \nmid d_x$, then $3d_Y > 4d_x$, which forces $3d_Y = 2d_z$, so $d_Y = 2\nu$ and $d_z = 3\nu$ for some integer $\nu \geq \lceil (2d_x + 1)/3 \rceil$.

(ii) ***Galois trace and divisorial interpolation:*** *Assume $(x(t), Y(t), z(t))$ is injective on a dense open subset of $\mathbb{A}^1$. Let $t_1, \ldots, t_{d_x}$ be the roots of $x(t) = x$ in $\overline{\mathbb{Q}(x)}$ and $P_i = (Y(t_i), z(t_i)) \in E(\overline{\mathbb{Q}(x)})$ the corresponding points on the generic fibre $E_x : z^2 = Y^3 - (x^4 + 1)$. Then*

$$P_1 + P_2 + \cdots + P_{d_x} = O \quad \text{in } E(\mathbb{Q}(x)). \tag{3}$$

Consequently, there exist unique polynomials $A(x, Y), B(x, Y) \in \mathbb{Q}[x, Y]$ with $\deg_Y A \leq \lfloor d_x/2 \rfloor$, $\deg_Y B \leq \lfloor (d_x - 3)/2 \rfloor$, and $\gcd_{\mathbb{Q}(x)[Y]}(A, B) = 1$ such that the divisor of zeroes of $A(x, Y) - B(x, Y)z$ on E_x is precisely $\sum_{i=1}^{d_x} [P_i]$, and the characteristic polynomial of $Y(t)$ over $\mathbb{Q}(x)$ is given up to a factor in $\mathbb{Q}(x)^*$ by the quotient

$$G_{d_x}(x, Y) = \frac{A(x, Y)^2 - B(x, Y)^2 (Y^3 - x^4 - 1)}{H(x, Y)}, \tag{4}$$

where $H(x, Y) \in \mathbb{Q}[x, Y]$ has $\deg_Y H = 2 \deg_Y B + 3 - d_x \in \{0, 1, 2\}$. In particular, when $B(x, Y) \in \mathbb{Q}[x]$ is independent of Y (which always holds for $d_x = 3$ and $d_x = 4$ whenever $P_1, \ldots, P_{d_x}$ are in general position), $z(t)$ admits the polynomial representation $z(t) = \frac{A(x(t), Y(t))}{B(x(t))}$.

Proof. (i) Write $x(t) = x_{d_x} t^{d_x} + \ldots$ and $z(t) = z_{d_z} t^{d_z} + \ldots$ with $x_{d_x} \in \mathbb{R}^*$ and $z_{d_z} \in \mathbb{R}$. In $Y(t)^3 = x(t)^4 + z(t)^2 + 1$, the leading coefficient of $x(t)^4$ is $x_{d_x}^4 > 0$ and that of $z(t)^2$ is $z_{d_z}^2 \geq 0$. Since both are non-negative and $x_{d_x}^4 > 0$, the highest-degree term of $x(t)^4 + z(t)^2 + 1$ has degree $\max(4d_x, 2d_z)$ and strictly positive coefficient $x_{d_x}^4$, $z_{d_z}^2$, or $x_{d_x}^4 + z_{d_z}^2$. Thus $3d_Y = \max(4d_x, 2d_z) \geq 4d_x$, giving $d_Y \geq \lceil 4d_x/3 \rceil$. If $3 \nmid d_x$, then $3d_Y \neq 4d_x$, so $3d_Y = 2d_z > 4d_x$, whence $d_Y = 2\nu$ and $d_z = 3\nu$.

(ii) The zero-cycle $\sum_{i=1}^{d_x} [P_i]$ on $E_{\mathbb{Q}(x)}$ is invariant under $\mathrm{Gal}(\overline{\mathbb{Q}(x)}/\mathbb{Q}(x))$, so its Abel–Jacobi sum lies in $E(\mathbb{Q}(x)) = \mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = \{O\}$ by Proposition 2.3 and Corollary 2.4 of [3]. By the Riemann–Roch theorem on the elliptic curve $E_{\mathbb{Q}(x)}$, the linear system $\mathcal{L}(d_x O)$ has dimension d_x with basis $\{1, Y, z, Y^2, Yz, \ldots\}$ of pole orders $\{0, 2, 3, 4, 5, \ldots, d_x\}$ at O. Since $\sum_{i=1}^{d_x} [P_i] \sim d_x O$ as divisors on $E_{\mathbb{Q}(x)}$, there is a unique function (up to $\mathbb{Q}(x)^*$-scaling) $f = A(x, Y) - B(x, Y)z \in \mathcal{L}(d_x O)$ vanishing on $\sum_{i=1}^{d_x} [P_i]$ (or a minimal sub-sum if a proper subset already sums to O). Taking norms from $\mathbb{Q}(x)[Y, z]/(z^2 - Y^3 + x^4 + 1)$ to $\mathbb{Q}(x)[Y]$ yields $\mathrm{N}(f) = A(x, Y)^2 - B(x, Y)^2 (Y^3 - x^4 - 1)$, whose divisor of zeroes on E_x is $\sum_{i=1}^{d_x} ([P_i] + [-P_i]) + R$, where R consists of at most two auxiliary points when $\max(2 \deg_Y A, 2 \deg_Y B + 3) > d_x$, projecting to $H(x, Y)$ of degree ≤ 2 in Y. □

Corollary 2.2 (Exclusion of Pullback and Unramified Families for Arbitrary $d_x \geq 3$)**.** *For any $d_x = \deg x \geq 3$, there is no nonconstant polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) satisfying any of the following:*

(a) *$(x(t), y(t), z(t)) = (\tilde{x}(q(t)), \tilde{y}(q(t)), \tilde{z}(q(t)))$ factors through a polynomial $q(t) \in \mathbb{Q}[t]$ with $\deg \tilde{x} \in \{1, 2\}$, or with $\deg \tilde{x} = 3$ and $\deg \tilde{y} \leq 8$;*

(b) *$Y(t) = Y_0(x(t))$ lies in $\mathbb{Q}[x(t)]$;*

(c) $z(t) = z_0(x(t))$ *lies in* $\mathbb{Q}[x(t)]$.

Proof. Part (a) follows immediately by substituting $s = q(t)$ and applying Theorem A ($\deg \tilde{x} \le 2$), Theorem B ($\deg \tilde{x} = 3, \deg \tilde{y} \le 4$), and Theorem 4.2 below ($\deg \tilde{x} = 3, \deg \tilde{y} = 8$). For (b), if $Y(t) = Y_0(x(t)) \in \mathbb{Q}[x]$, then $z(t)^2 = Y_0(x(t))^3 - x(t)^4 - 1 = F(x(t))$. By Lemma 5.1 of [3], the d_x points $P_i = (Y_0(x), z(t_i))$ have the same Y-coordinate, so $z(t_i) = \pm\sqrt{F(x)}$. If d_x is odd, then $P_1 + \cdots + P_{d_x} = (d_x - 2m)P_1 = O$ with $d_x - 2m \neq 0$, forcing $P_1 \in E(\mathbb{Q}(x))[d_x - 2m] = \{O\}$ (since $\mathrm{MW}(\hat{E}/\mathbb{Q}(x)) = 0$), impossible. If $d_x = 2m$ is even, then $z(t)$ must be invariant under the pairs (t_i, t_{i+m}) with $z(t_{i+m}) = -z(t_i)$, reducing to a quadratic extension of $\mathbb{Q}(x)$ which is excluded by Theorem A. For (c), if $z(t) = z_0(x(t)) \in \mathbb{Q}[x]$, then $Y(t)^3 = x(t)^4 + z_0(x(t))^2 + 1 = f(x(t))$ with $f(x) \ge 1$ on $\mathbb{R}$. Since $f(x)$ has no real roots, every root $\rho \in \mathbb{C}$ of $f(x)$ with multiplicity not divisible by 3 must be a root of multiplicity divisible by 3 of $x(t) - \rho$; if $3 \nmid \deg_x f$, at least one root ρ has multiplicity in f not divisible by 3, which forces $x(t) - \rho$ to have a multiple root in $\mathbb{C}$, and if $\deg_x f = 4$ (when $\deg_x z_0 \le 2$), the three critical values of $x(t)$ cannot cover four distinct roots of $f(x)$ in conjugate pairs without a real root, contradicting $f \ge 1$ on $\mathbb{R}$. □

2.2 Universal Upper Bounds for Arbitrary $\deg x \ge 3$

We now prove that for *every* fixed degree $d_x = \deg x \ge 3$, there are only finitely many possible degree pairs (d_Y, d_z) for a polynomial solution $(x(t), Y(t), z(t))$ (even over $\mathbb{C}$).

Theorem 2.3 (Universal Degree Bound for Arbitrary $\deg x \ge 3$)**.** *Let $(x(t), Y(t), z(t)) \in \mathbb{C}[t]^3$ be a nonconstant polynomial solution to $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ with $d_x = \deg x \ge 3$, $d_Y = \deg Y$, and $d_z = \deg z$. Assume that $Y(t)$ and $z(t)$ are coprime in $\mathbb{C}[t]$ (which holds automatically over $\mathbb{Q}$ whenever every critical value of $x(t)$ has degree ≤ 3 over $\mathbb{Q}$, and in particular for $d_x \in \{3, 4\}$). Then (d_Y, d_z) falls into one of three finite regimes:*

(1) ***Sub-canonical regime*** *($3d_Y < 4d_x$,* ***over*** *$\mathbb{C} \setminus \mathbb{R}$* ***only*****):** *$d_z = 2d_x$ and $1 \le d_Y \le \lfloor (4d_x - 1)/3 \rfloor$. By Theorem 2.1(i), this regime contains* ***no solutions over*** *$\mathbb{R}$* ***or*** *$\mathbb{Q}$.*

(2) ***Canonical regime*** *($3d_Y = 4d_x$)**:** $d_x = 3k$ *is a multiple of* 3, $d_Y = 4k$, *and* $d_z \le 6k$ *(with $d_z = 6k$ over $\mathbb{R}$ or $\mathbb{Q}$).*

(3) ***Hall–Davenport regime*** *($3d_Y = 2d_z > 4d_x$)**:** $d_Y = 2\nu$ *is even and* $d_z = 3\nu$ *for an integer ν satisfying the universal bounds*

$$\left\lfloor \frac{2d_x}{3} \right\rfloor + 1 \le \nu \le 4d_x - 1, \qquad \text{so} \qquad \frac{4d_x + 2}{3} \le d_Y \le 8d_x - 2. \tag{5}$$

Without assuming $\gcd(Y(t), z(t)) = 1$, *let* $g_{\mathrm{cd}} = \deg \gcd(Y(t), z(t))$. *Then* $g_{\mathrm{cd}} \le d_x - 1$ *(since any root of* $\gcd(Y, z)$ *is a multiple root of* $x(t)^4 + 1$, *hence a root of* $x'(t)$*), and the Mason–Stothers theorem gives* $\nu \le 5d_x - 2$ *($d_Y \le 10d_x - 4$) unconditionally for all* $d_x \ge 3$.

Proof. If $3d_Y < 4d_x$, then in $Y(t)^3 - z(t)^2 = x(t)^4 + 1$ the leading terms of $-z(t)^2$ and $x(t)^4$ must match, so $2d_z = 4d_x \implies d_z = 2d_x$ (which requires $z_{d_z}^2 = -x_{d_x}^4$, impossible over $\mathbb{R}$). If $3d_Y = 4d_x$, then $3 \mid d_x$, so $d_x = 3k$ and $d_Y = 4k$.

Now suppose $3d_Y > 4d_x$. Then the leading terms of $Y(t)^3$ and $z(t)^2$ must cancel, so $3d_Y = 2d_z$, which forces $d_Y = 2\nu$ and $d_z = 3\nu$ with $6\nu > 4d_x$, i.e., $\nu \ge \lfloor 2d_x/3 \rfloor + 1$. When $\gcd(Y, z) = 1$, $Y(t)$ and $z(t)$ are coprime polynomials with $Y(t)^3 - z(t)^2 \neq 0$, so Davenport's inequality [6, 7, 9] yields

$$4d_x = \deg\left(x(t)^4 + 1\right) = \deg\left(Y(t)^3 - z(t)^2\right) \ge \frac{1}{2}\deg Y + 1 = \nu + 1,$$

whence $\nu \le 4d_x - 1$ and $d_Y = 2\nu \le 8d_x - 2$.

In the general case where $D_0(t) = \gcd(Y(t), z(t))$ has degree $g_{\mathrm{cd}} \ge 0$, every root $\alpha \in \mathbb{C}$ of $D_0(t)$ divides $Y(t)^3$ to order ≥ 3 and $z(t)^2$ to order ≥ 2, so α is a root of $x(t)^4 + 1$ of multiplicity $m_\alpha \ge 2$.

Since X^4+1 is separable, $x(t)-\zeta_8^{2j+1}$ has multiplicity $m_\alpha \geq 2$ at α, so $x'(t)$ vanishes at α to order $m_\alpha - 1 \geq \frac{1}{2}m_\alpha$. Summing over all roots of $D_0(t)$ gives $g_{\mathrm{cd}} \leq \deg x'(t) = d_x - 1$. Applying the Mason–Stothers inequality [8] to the coprime triple $(Y^3/D_0^2,\ -z^2/D_0^2,\ -(x^4+1)/D_0^2)$ gives

$$6\nu - 2g_{\mathrm{cd}} \leq N_0(Y) + N_0(z) + N_0(x^4+1) - g_{\mathrm{cd}} - 1 \leq 2\nu + 3\nu + 4d_x - g_{\mathrm{cd}} - 1,$$

so $\nu \leq 4d_x + g_{\mathrm{cd}} - 1 \leq 5d_x - 2$. Finally, for $d_x \in \{3,4\}$ over $\mathbb{Q}$, any root of $x'(t)$ has degree $\leq d_x - 1 \leq 3$ over $\mathbb{Q}$, so $x(\alpha)$ has degree ≤ 3 over $\mathbb{Q}$ and cannot satisfy $x(\alpha)^4+1=0$ (as $\Phi_8(X) = X^4+1$ is irreducible of degree 4 over $\mathbb{Q}$). Thus $g_{\mathrm{cd}} = 0$ holds unconditionally over $\mathbb{Q}$ for $d_x \in \{3,4\}$. □

Universal Finiteness of Degree Patterns for $\deg x \geq 3$: For every degree $d_x = \deg x \geq 3$, Theorem 2.3 confines any rational polynomial family on $x^4+y^3+z^2+1=0$ to the finite range $\lceil 4d_x/3 \rceil \leq \deg y \leq 8d_x - 2$ (for $d_x \in \{3,4\}$) and $\deg y \leq 10d_x - 4$ (for all $d_x \geq 5$). For $d_x = 4$, this restricts $(\deg y, \deg z)$ to $(2\nu, 3\nu)$ with $3 \leq \nu \leq 15$ ($\deg y \in \{6, 8, \ldots, 30\}$).

3 Binary Cubic Covariants and Bihomogeneous Torus Reduction for $\deg x = 3$

We now specialize to $\deg x = 3$ and arbitrary $\deg Y = d = 2\nu \geq 4$. Over $\overline{\mathbb{Q}}$ (or over $\mathbb{Q}$ up to affine change of t and scaling), normalise $x(t) = t^3 + st + x_0$ and put $u = x - x_0 = t^3 + st$. By Lemma 6.1 of [3], any polynomial solution with $\deg x = 3$ satisfies

$$z(t) = m(u)Y(t) + n(u), \qquad m(u), n(u) \in \mathbb{Q}[u], \quad m(u) \neq 0, \tag{6}$$

and $Y(t)$ has characteristic polynomial $G(x,Y) = Y^3 - m(u)^2Y^2 - 2m(u)n(u)Y - (n(u)^2 + (u+x_0)^4 + 1) = 0$ over $\mathbb{Q}(u)$. Writing $Y(t) = \alpha(u) + \beta(u)t + \gamma(u)t^2$ in the $\mathbb{Q}[u]$-basis $\{1, t, t^2\}$ of $\mathbb{Q}[t]$, the trace condition $\mathrm{Tr}_{\mathbb{Q}(t)/\mathbb{Q}(u)}(Y) = 3\alpha - 2s\gamma = m^2$ gives

$$Y(t) = \frac{1}{3}m(u)^2 + W(t), \qquad W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2, \qquad \mathrm{Tr}(W) = 0. \tag{7}$$

Theorem 3.1 (Binary Cubic Covariant Structure and Unified Bihomogeneous Identity)**.** *Let $W(t) = \frac{2}{3}s\gamma(u) + \beta(u)t + \gamma(u)t^2 \in \mathbb{Q}[u][t]/(t^3 + st - u)$, and let $T^3 + p(u)T - N(u)$ be the characteristic polynomial of W over $\mathbb{Q}[u]$.*

(i) ***Covariant representation and discriminant syzygy:*** *Let $I(\beta,\gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$ be the index cubic of W with respect to the basis $\{1,t,t^2\}$. Then $p(u)$ and $N(u)$ are the classical Hessian and Jacobian covariants of the binary cubic form $I(\beta,\gamma)$:*

$$p(u) = \frac{1}{12}\,\mathrm{Hess}_{\beta,\gamma}(I) = s\beta^2 - 3u\beta\gamma - \frac{1}{3}s^2\gamma^2, \tag{8}$$

$$N(u) = -\frac{1}{9}\,\mathrm{Jac}_{\beta,\gamma}(I,p) = u\beta^3 + \frac{2}{3}s^2\beta^2\gamma - su\beta\gamma^2 + \left(u^2 + \frac{2}{27}s^3\right)\gamma^3, \tag{9}$$

and they satisfy the universal covariant syzygy

$$4p(u)^3 + 27N(u)^2 = (27u^2 + 4s^3)\,I(\beta(u),\gamma(u))^2. \tag{10}$$

(ii) ***Unified polynomial identity:*** *A quadruple $(s, x_0, \beta(u), \gamma(u), m(u))$ with $m(u) \neq 0$ defines a polynomial solution $(x(t), Y(t), z(t))$ to (1) if and only if it satisfies the single polynomial identity in $\mathbb{Q}[u]$:*

$$\Phi(u) := \Psi(u) - 4m(u)^2\big((u+x_0)^4 + 1\big) \equiv 0, \tag{11}$$

where

$$\Psi(u) := 4m(u)^2N(u) - p(u)^2 + \frac{2}{3}m(u)^4p(u) + \frac{1}{27}m(u)^8. \tag{12}$$

Whenever (11) holds, $m(u)^2$ divides $p(u)^2$ in $\mathbb{Q}[u]$, so $m(u)$ divides $p(u)$ automatically and $n(u) := -\frac{p(u)}{2m(u)} - \frac{1}{6}m(u)^3$ is a polynomial in $\mathbb{Q}[u]$.

(iii) ***Exact degree bounds and $\mathbb{G}_{\mathrm{m}}^2$-bihomogeneity:*** *For $d = \deg_t Y(t) = 2\nu \in \{8, 10, 14, 16, 20, 22\}$, we have*

$$\deg_u \beta \le \left\lfloor \frac{2\nu - 1}{3} \right\rfloor, \quad \deg_u \gamma \le \left\lfloor \frac{2\nu - 2}{3} \right\rfloor, \quad \mu := \deg_u m \le \left\lfloor \frac{\nu}{3} \right\rfloor \le \frac{\nu - 1}{3}, \quad \deg_u n = \nu. \tag{13}$$

In particular, $\deg_u(m^3) = 3\mu \le \nu - 1 < \nu = \deg_u n$ for all six open cases. Furthermore, $\Psi(u)$ is bihomogeneous under the two-dimensional torus $\mathbb{G}_{\mathrm{m}}^2 = \{(\lambda, \mu) \in (\overline{\mathbb{Q}}^*)^2\}$ acting via

$$(u, s, x_0) \mapsto (\lambda^3 u, \lambda^2 s, \lambda^3 x_0), \qquad (\beta, \gamma, m, n) \mapsto (\lambda^{-1}\mu^2\beta,\ \lambda^{-2}\mu^2\gamma,\ \mu m,\ \mu^3 n), \tag{14}$$

under which $\Psi(u) \mapsto \mu^8 \Psi(u)$.

Proof. (i) Multiplication by $W = \frac{2}{3}s\gamma + \beta t + \gamma t^2$ on the $\mathbb{Q}[u]$-basis $(1, t, t^2)$ of $\mathbb{Q}[u][t]/(t^3 + st - u)$ has matrix

$$M_W = \begin{pmatrix} \frac{2}{3}s\gamma & u\gamma & u\beta \\ \beta & -\frac{1}{3}s\gamma & u\gamma - s\beta \\ \gamma & \beta & -\frac{1}{3}s\gamma \end{pmatrix},$$

whose characteristic polynomial $\det(TI_3 - M_W) = T^3 + pT - N$ gives (8) and (9). Direct differentiation of $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$ gives $\partial_\beta^2 I \cdot \partial_\gamma^2 I - (\partial_\beta \partial_\gamma I)^2 = 12p$ and $\partial_\beta I \cdot \partial_\gamma p - \partial_\gamma I \cdot \partial_\beta p = -9N$. Since $\mathrm{Disc}_t(t^3 + st - u) = -(27u^2 + 4s^3)$ and $[1, t, t^2]$ vs. $[1, W, W^2]$ has index determinant $\det(1, W, W^2) = I(\beta, \gamma)$, the discriminant identity $\mathrm{Disc}_T(T^3 + pT - N) = -(4p^3 + 27N^2) = -(27u^2 + 4s^3)I^2$ yields (10).

(ii) Shifting $Y = W + \frac{1}{3}m^2$ in $G(x, Y) = Y^3 - m^2Y^2 - 2mnY - (n^2 + x^4 + 1)$ gives

$$W^3 - 2m\left(n + \frac{1}{6}m^3\right)W - \left(\left(n + \frac{1}{3}m^3\right)^2 - \frac{1}{27}m^6 + x^4 + 1\right) = 0.$$

Comparing with $W^3 + pW - N = 0$ yields $p = -2m(n + \frac{1}{6}m^3)$ and $N = (n + \frac{1}{3}m^3)^2 - \frac{1}{27}m^6 + x^4 + 1$. Writing $q := n + \frac{1}{6}m^3 = -\frac{p}{2m}$, we have $n + \frac{1}{3}m^3 = q + \frac{1}{6}m^3$, so $(n + \frac{1}{3}m^3)^2 - \frac{1}{27}m^6 = q^2 + \frac{1}{3}m^3 q - \frac{1}{108}m^6$. Multiplying $N - (q^2 + \frac{1}{3}m^3 q - \frac{1}{108}m^6) = x^4 + 1$ by $4m^2$ and substituting $2mq = -p$ gives (11). Conversely, if (11) holds in $\mathbb{Q}[u]$, every term except $-p(u)^2$ is divisible by $m(u)^2$, so $m(u)^2 \mid p(u)^2 \implies m(u) \mid p(u)$, making $n(u)$ a polynomial.

(iii) Since $\deg_t(\alpha(u)) \equiv 0$, $\deg_t(\beta(u)t) \equiv 1$, and $\deg_t(\gamma(u)t^2) \equiv 2 \pmod 3$, there is no leading-term cancellation in $Y(t) = \alpha(u) + \beta(u)t + \gamma(u)t^2$. Thus $3\deg_u \alpha \le 2\nu \implies 2\deg_u m = \deg_u(m^2) \le \lfloor 2\nu/3 \rfloor \implies \deg_u m \le \lfloor \nu/3 \rfloor$, and $3\deg_u \beta + 1 \le 2\nu$, $3\deg_u \gamma + 2 \le 2\nu$ yield (13). Since $3 \nmid \nu$ for $\nu \in \{4, 5, 7, 8, 10, 11\}$, $\lfloor \nu/3 \rfloor \le (\nu - 1)/3$, so $3\mu \le \nu - 1 < \nu$. Finally, under (14), $p \mapsto \mu^4 p$ and $N \mapsto \mu^6 N$, so $\Psi \mapsto \mu^8 \Psi$. □

Remark 3.2 (Why the Section 7.1 Gröbner basis in [3] failed to terminate). In Section 7.1 of [3], the system for $\nu = 4$ ($d = 8$) was formulated using $4m^2e_3 - e_2^2 - 4m^2(x^4 + 1) = 0$ without quotienting out the two-dimensional torus action (14) on the homogeneous equations $u^{10}, \ldots, u^1$. Because the top equations are invariant under $\mathbb{G}_{\mathrm{m}}^2$ while the single u^0 equation breaks the symmetry via $+1$, unreduced Buchberger algorithms suffer severe degree blow-up. By quotienting out $\mathbb{G}_{\mathrm{m}}^2$ and exploiting the triangular linearity of $\Psi(u)$ in the subleading coefficients of m, β, γ, the system reduces to bivariate/univariate resultants that execute in seconds.

4 Complete Resolution of the Open $\nu = 4$ ($\deg x = 3, \deg y = 8, \deg z = 12$) System

For $\nu = 4$ ($d = \deg_t Y = 8, \deg_t z = 12$), Theorem 3.1(iii) gives $\deg_u \beta \le 2$, $\deg_u \gamma = 2$, and $\deg_u m \le 1$. Write

$$\beta(u) = b_2u^2 + b_1u + b_0, \qquad \gamma(u) = g_2u^2 + g_1u + g_0 \quad (g_2 \neq 0), \qquad m(u) = m_1u + m_0. \tag{15}$$

The leading coefficient of $\Phi(u)$ in (11) at u^{10} is

$$\mathrm{coeff}_{u^{10}}(\Phi) = -g_2^2(9b_2^2 - 4g_2m_1^2) = 0 \implies 9b_2^2 = 4g_2m_1^2. \tag{16}$$

Since $g_2 \neq 0$, this splits cleanly into two branches: $m_1 = 0$ (which forces $b_2 = 0$) and $m_1 \neq 0$ (which forces $b_2 \neq 0$ and $g_2 = \frac{9b_2^2}{4m_1^2}$).

4.1 The Constant-m Branch ($m_1 = 0$) and Discovery of the Explicit $(3, 8, 12)$ Curve over $\overline{\mathbb{Q}}$

Theorem 4.1 (Complete Solution of the $m_1 = 0$ Branch for $\nu = 4$)**.** *In the case* $\deg x = 3, \deg Y = 8, \deg z = 12$ *with* $m(u) = m_0 \in \overline{\mathbb{Q}}^*$ *constant (so* $b_2 = 0$*):*

(i) ***No rational solutions:*** *There is no solution* $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ *with* $m_1 = 0$.

(ii) ***Explicit solution over*** $\mathbb{Q}(\sqrt{3}, 2^{1/3})$***:*** *Over* $\overline{\mathbb{Q}}$*, the* $m_1 = 0$ *branch is* ***non-empty****. Setting* $s = 0, x_0 = 0, b_1 = -2, g_2 = (27/4)^{1/3} = \frac{3}{2}2^{1/3}, m_0 = -2^{1/3}\sqrt{3}$ *yields the explicit polynomial identity of degrees* $(3, 8, 12)$*:*

$$\begin{aligned} x(t) &= t^3, \\ y(t) &= -\frac{3}{2}\,2^{1/3}t^8 + 2t^4 - 2^{2/3}, \\ z(t) &= \sqrt{3}\left(\frac{3}{2}\,t^{12} - \frac{3}{2}\,2^{2/3}t^8 + 2\cdot 2^{1/3}t^4 - 1\right), \end{aligned} \tag{17}$$

which satisfies $x(t)^4 + y(t)^3 + z(t)^2 + 1 = 0$ *identically in* $\mathbb{Q}(\sqrt{3}, 2^{1/3})[t]$*. In particular, the moduli space of polynomial curves with* $\deg x = 3, \deg y = 8, \deg z = 12$ *is non-empty over* $\overline{\mathbb{Q}}$*, disproving the heuristic expectation in Appendix C of [3].*

Proof. With $m_1 = 0$ and $b_2 = 0$, $\Phi(u)$ has degree 8 in u. Its u^8 coefficient is

$$\mathrm{coeff}_{u^8}(\Phi) = -g_2^2\Big((9b_1 + g_2s^2)^2 - 36g_2m_0^2\Big) = 0 \implies m_0^2 = \frac{(9b_1 + g_2s^2)^2}{36g_2}.$$

Since $m_0 \neq 0$ and $g_2 \neq 0$, we have $9b_1 + g_2s^2 \neq 0$. Substituting m_0^2 into $\Phi(u)$, the coefficients of u^7 and u^6 are linear in b_0 and g_0 respectively:

$$\mathrm{coeff}_{u^7}(\Phi) = -\frac{g_2(9b_1 + g_2s^2)}{9}(18b_0g_2 - g_1(9b_1 - g_2s^2)) = 0 \implies b_0 = \frac{g_1(9b_1 - g_2s^2)}{18g_2}, \tag{18}$$

$$\mathrm{coeff}_{u^6}(\Phi) = 0 \implies g_0 = \frac{324b_1^2g_2s + 243b_1g_1^2 + 36b_1g_2^2s^3 + 27g_1^2g_2s^2 - 8g_2^3s^5}{108g_2(9b_1 - g_2s^2)}. \tag{19}$$

(If $9b_1 - g_2s^2 = 0$, then $b_1 = \frac{1}{9}g_2s^2 \neq 0$ and $\mathrm{coeff}_{u^6}(\Phi)$ becomes $-\frac{1}{27}g_2^2s^2(9g_1^2 + g_2^2s^3) = 0$, which upon substitution into u^5 and u^4 yields $g_2s = 0$, a contradiction.) Substituting (18)–(19) into $\mathrm{coeff}_{u^5}(\Phi) = 0$ gives

$$\frac{g_1s(9b_1 + g_2s^2)}{324(9b_1 - g_2s^2)}\Big(9g_1^2s(9b_1 + g_2s^2) + 2(9b_1 - g_2s^2)^2(3b_1 + g_2s^2)\Big) = 0. \tag{20}$$

This splits into three subcases:

Subcase 1 ($s = 0$): When $s = 0$, $b_1 \neq 0$ and the remaining coefficients of $\Phi(u)$ simplify to

$$u^4: \quad -\frac{9b_1^2(b_1^3 + 8)}{8g_2} = 0, \qquad u^3: \quad -\frac{9b_1^2(3b_1^3g_1 + 64g_2x_0)}{16g_2^2} = 0,$$

$$u^2: \quad -\frac{27b_1^2(b_1^3g_1^2 + 64g_2^2x_0^2)}{32g_2^3} = 0, \qquad u^0: \quad -\frac{9b_1^2(-27b_1^6 + 256g_2^3(x_0^4 + 1))}{256g_2^4} = 0.$$

From u^4, $b_1^3 = -8$ (so $b_1 = -2$ over $\mathbb{Q}$). Then u^3 gives $64g_2x_0 = -3b_1^3g_1 = 24g_1$, while u^2 gives $64g_2^2x_0^2 = -b_1^3g_1^2 = 8g_1^2$. Comparing $(64g_2x_0)^2 = 576g_1^2$ with $64(64g_2^2x_0^2) = 512g_1^2$ forces $g_1 = 0$ and $x_0 = 0$. Finally, u^0 gives $256g_2^3 = 27b_1^6 = 1728$, so

$$g_2^3 = \frac{27}{4}.$$

Since 4 is not a cube in $\mathbb{Q}$, there is no solution in $\mathbb{Q}[t]^3$. Over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$, taking $g_2 = \frac{3}{2}2^{1/3}$, $b_1 = -2$, $m_0 = -2^{1/3}\sqrt{3}$ (so $m_0^2 = 9/g_2 = 3 \cdot 2^{2/3}$) and $n(u) = -\frac{p(u)}{2m_0} - \frac{1}{6}m_0^3 = \frac{3\sqrt{3}}{2}u^4 + \frac{\sqrt{3}}{2}$ yields (17). Direct expansion gives

$$Y(t)^3 = \frac{27}{4}t^{24} - 27 \cdot 2^{2/3}t^{20} + 4\left(\frac{27}{2} \cdot 2^{1/3} + 9 \cdot 2^{1/3}\right)t^{16} - (8+36)t^{12} + 36 \cdot 2^{2/3}t^8 - 12 \cdot 2^{1/3}t^4 + 4,$$

and $z(t)^2 = Y(t)^3 - t^{12} - 1$, confirming $Y(t)^3 - z(t)^2 = x(t)^4 + 1$.

Subcase 2 ($g_1 = 0, s \neq 0$): When $g_1 = 0$, $\mathrm{coeff}_{u^3}(\Phi) = -\frac{4x_0(9b_1+g_2s^2)^2}{9g_2} = 0$ forces $x_0 = 0$. Subtracting $\frac{9}{2s^3}\mathrm{coeff}_{u^2}(\Phi)$ from $\mathrm{coeff}_{u^4}(\Phi)$ then leaves $-\frac{(9b_1+g_2s^2)^2}{9g_2} = 0$, contradicting $9b_1 + g_2s^2 \neq 0$.

Subcase 3 ($g_1 s \neq 0$): Writing $b_1 = \lambda g_2 s^2$ and $W_6 = g_2^3 s^6$, equation (20) gives $g_1^2 = -\frac{2g_2^2s^3(3\lambda+1)(9\lambda-1)^2}{9(9\lambda+1)}$. Substituting this into $\mathrm{coeff}_{u^4}(\Phi) = 0$ and $\mathrm{coeff}_{u^3}(\Phi) = 0$ solves linearly for W_6 and x_0:

$$W_6 = -\frac{1944(9\lambda+1)^2}{(3\lambda+1)(9\lambda-1)^2(81\lambda^2+54\lambda+25)}.$$

Substituting g_1^2, W_6, x_0 into $\mathrm{coeff}_{u^2}(\Phi) = 0$ and $\mathrm{coeff}_{u^1}(\Phi) = 0$ yields two univariate polynomial conditions on λ:

$$\begin{aligned} P_2(\lambda) &= 14348907\lambda^7 + 1594323\lambda^6 - 9743085\lambda^5 - 12052557\lambda^4 \\ &\quad - 4758183\lambda^3 - 631071\lambda^2 + 389673\lambda + 104089 = 0, \\ P_1(\lambda) &= 1420541793\lambda^9 + 1018772397\lambda^8 - 752520456\lambda^7 - 2426205312\lambda^6 - 2183592654\lambda^5 \\ &\quad - 1128723822\lambda^4 - 287583696\lambda^3 - 13530456\lambda^2 + 17051205\lambda + 3074969 = 0. \end{aligned}$$

In $\mathbb{Q}[\lambda]$, $\gcd(P_1(\lambda), P_2(\lambda)) = 1$, so no solution exists even over $\mathbb{C}$. □

4.2 The Linear-m Branch ($m_1 \neq 0$) for $\nu = 4$

Theorem 4.2 (Nonexistence of Solutions in the $m_1 \neq 0$ Branch for $\nu = 4$)**.** *There is no polynomial solution $(x(t), y(t), z(t)) \in \mathbb{Q}[t]^3$ to (1) with $\deg x = 3$ and $\deg y = 8$ ($\deg z = 12$).*

Proof. By Theorem 4.1, it remains to exclude the branch $m_1 \neq 0$, where $b_2 \neq 0$ and $g_2 = \frac{9b_2^2}{4m_1^2}$ by (16). Using the two-dimensional torus action (14) of Theorem 3.1(iii) with $\lambda = b_2^{1/2}m_1^{-1}$ and $\mu = b_2^{1/2}m_1^{-2}$, we pass to dimensionless variables by setting $b_2 = 1$, $m_1 = 1$, $g_2 = \frac{9}{4}$, so that $\Phi(u) = \Psi(u) - 4W(u+m_0)^2(u+x_0)^4 - 4W'(u+m_0)^2 \equiv 0$, where $W = b_2^6/m_1^{18} \in \mathbb{Q}^*$.

The coefficients of u^9, u^8, u^7 in $\Psi(u)$ are triangular and linear in m_0, b_0, g_0 respectively:

$$\mathrm{coeff}_{u^9}(\Psi) = 0 \implies m_0 = b_1 - \frac{2}{9}g_1 + \frac{1}{4}s^2 + \frac{2}{27}s + \frac{4}{729}, \tag{21}$$

$$\begin{aligned} \mathrm{coeff}_{u^8}(\Psi) = 0 \implies b_0 &= \frac{2}{9}b_1g_1 - \frac{4}{27}b_1s - \frac{16}{729}b_1 + \frac{2}{9}g_0 - \frac{2}{27}g_1^2 - \frac{1}{18}g_1s^2 + \frac{4}{81}g_1s + \frac{56}{6561}g_1 \\ &\quad - \frac{1}{54}s^3 - \frac{10}{729}s^2 - \frac{56}{19683}s - \frac{32}{177147}. \end{aligned} \tag{22}$$

Substituting (21)–(22) into $\mathrm{coeff}_{u^7}(\Psi)$, the coefficient of g_0 is $D := \frac{81}{4}b_1 - 9g_1 - \frac{81}{16}s^2 + \frac{3}{2}s + \frac{1}{3}$. Introducing the linear coordinate change $(b_1, g_1, s) \mapsto (\tilde{D}, E, S)$ defined by

$$s = \frac{S-4}{27}, \qquad g_1 = \frac{E-S+8}{81}, \qquad b_1 = \frac{\tilde{D}}{243} + \frac{4g_1}{9} + \frac{s^2}{4} - \frac{2s}{27} - \frac{4}{243} \quad (\text{so } D = \tilde{D}/12), \tag{23}$$

$\mathrm{coeff}_{u^7}(\Psi) = 0$ solves linearly for g_0 with the *monomial* denominator $\tilde{D}$:

$$g_0 = \frac{18\tilde{D}^2(S+8) + 6\tilde{D}(E^2 + 2ES + 16E + 9S^3 - 36S^2 - 192S + 768) + E^2S^2}{354294\tilde{D}}. \tag{24}$$

(If $\tilde{D} = 0$, then $\mathrm{coeff}_{u^7}(\Psi) = -E^2S^2/4251528 = 0$ forces $E = 0$ or $S = 0$, and in both cases $\mathrm{coeff}_{u^6}(\Psi) = -\square \leq 0$ conflicts with $W = b_2^6/m_1^{18} > 0$ over $\mathbb{Q}$; indeed even over $\mathbb{C}$, $p(-m_0) = 0$ has no solution with $W \neq 0$.)

Next, since $m(u) = u + m_0$ divides $p(u)$, the remainder $p_{\mathrm{rem}} := p(-m_0)$ must vanish, and with $\tilde{q}(u) := -\frac{p(u)-p(-m_0)}{2(u+m_0)}$, the exact polynomial quotient

$$Q_{\mathrm{exact}}(u) := N(u) - \tilde{q}(u)^2 - \frac{1}{3}(u+m_0)^3\tilde{q}(u) + \frac{1}{108}(u+m_0)^6 = Q_4u^4 + Q_3u^3 + Q_2u^2 + Q_1u + Q_0$$

must satisfy $Q_{\mathrm{exact}}(u) = W(u+x_0)^4 + W'$. Eliminating W and x_0 from Q_4, Q_3, Q_2, Q_1 gives the two fourth-power invariants

$$3Q_3^2 - 8Q_4Q_2 = 0, \qquad Q_3Q_2 - 6Q_4Q_1 = 0, \qquad \text{together with} \qquad p(-m_0) = 0. \tag{25}$$

Applying the blow-up substitution $E = \frac{\tilde{D}f}{S}$ (for $S = 0$, $p(-m_0) = 0$ gives $E = -\frac{3\tilde{D}(9\tilde{D}+128)}{8(3\tilde{D}+32)}$, for which $3Q_3^2 - 8Q_4Q_2$ has no roots with $Q_4 \neq 0$), a factor of $\tilde{D}^2$ cancels from $p(-m_0), Q_4, Q_3$, and the degree of $p(-m_0)$ in $\tilde{D}$ drops from 6 to 2:

$$P_{\mathrm{rem}}(\tilde{D}, S, f) = A_0(S,f)\tilde{D}^2 + B_0(S,f)\tilde{D} + C_0(S,f) = 0, \tag{26}$$

where

$$\begin{aligned}
A_0(S,f) &= 108\big(18S(f+9) + f(f+12)^2\big),\\
B_0(S,f) &= 4\big(243S^3(f+8) + 9S^2(f^3 + 21f^2 - 36f - 1296) + 432S(f^2 + 24f + 144) + 2f(f+12)^4\big),\\
C_0(S,f) &= 3S^3(S-4)(f+6)(36S + f^2 + 24f).
\end{aligned}$$

Simultaneously, both Q_4 and Q_3 share the factor $(f+12)$ (corresponding to $12\tilde{D} + ES = 0$, which cannot vanish since $Q_4 = W \neq 0$), with Q_4 linear in $\tilde{D}$:

$$Q_4 = -\frac{\tilde{D}^2(f+12)\big(36(18S + f^2 + 12f)\tilde{D} + S(6S + f - 12)(36S + f^2 - 144)\big)}{223154201664S}.$$

Dividing $3Q_3^2 - 8Q_4Q_2$ and $Q_3Q_2 - 6Q_4Q_1$ by $\tilde{D}^3(f+12)$ yields polynomials $I_2(\tilde{D}, S, f)$ and $I_3(\tilde{D}, S, f)$ of degrees 5 and 6 in $\tilde{D}$. Pseudo-dividing I_2 and I_3 by the quadratic P_{rem} with respect to $\tilde{D}$ gives two linear relations $L_2 = A_{2,\mathrm{red}}(S,f)\tilde{D} + B_{2,\mathrm{red}}(S,f) = 0$ and $L_3 = A_{3,\mathrm{red}}(S,f)\tilde{D} + B_{3,\mathrm{red}}(S,f) = 0$ (of degrees 15 and 18 in S). Eliminating $\tilde{D}$ via $\mathrm{Res}_{\tilde{D}}(P_{\mathrm{rem}}, I_2) = 0$ and $A_{2,\mathrm{red}}B_{3,\mathrm{red}} - A_{3,\mathrm{red}}B_{2,\mathrm{red}} = 0$ factors out the boundary components $S^7(S-4)(f+6)(f+12)^2(36S + f^2 + 24f)A_0(S,f)^4$ and leaves two coprime irreducible bivariate polynomials $F_{12}(S,f) = 0$ and $F_{23}(S,f) = 0$, each of degree 20 in S and degrees 28 and 29 in f.

In both F_{12} and F_{23}, the coefficient of S^k is divisible by $(f+12)^{16-k}$ for $0 \leq k \leq 16$. Setting $S = W_S(f+12)$ and dividing by $(f+12)^{16}$ reduces $F_{12,1}(W_S, f)$ and $F_{23,1}(W_S, f)$ to degrees 16 and 17 in f. Their resultant $R_{12,23}(f) := \mathrm{Res}_{W_S}(F_{12,1}, F_{23,1}) \in \mathbb{Z}[f]$ has exact degree 530 and factors structurally as

$$R_{12,23}(f) = c \cdot f^{80}(f+6)^{40}(f+12)^{24}(f-12)^{20}(f^2 + 18f + 108)^{16}(f+9) \cdot H_{\mathrm{sp}}(f)^2 \cdot H_{\mathrm{true}}(f), \tag{27}$$

where:

- $f(f+6)(f+12)(f-12)(f^2+18f+108)(f+9)$ is the boundary locus; direct substitution of $f \in \{0,-6,12,-12,-9\}$ into (26) yields only $\tilde{D}=0$ or $S=0$;
- $H_{\mathrm{sp}}(f)$ of degree $108=17+91$ is the spurious locus $A_{2,\mathrm{red}}(W_S,f)=B_{2,\mathrm{red}}(W_S,f)=0$ (which appears with multiplicity 2 since $F_{12}\in(A_{2,\mathrm{red}},B_{2,\mathrm{red}})^2$ and $F_{23}\in(A_{2,\mathrm{red}},B_{2,\mathrm{red}})^1$), verified to have no linear factors modulo $p=100057$ (factor degrees $2,2,6,10,15,18,27,28$);
- $H_{\mathrm{true}}(f)$ of degree 117 is the true elimination polynomial of $(P_{\mathrm{rem}},I_2,I_3)$. Modulo the primes $p_1=10037$, $p_2=10067$, and $p_3=10133$ (all of good reduction, where $\deg R_{12,23}=530$), $H_{\mathrm{true}}(f)$ factors into irreducible polynomials of degrees

$$\begin{aligned} p_1=10037: &\quad [2,8,9,10,22,25,41],\\ p_2=10067: &\quad [2,3,6,18,19,69],\\ p_3=10133: &\quad [2,2,4,5,18,86]. \end{aligned}$$

  None of these factorizations contains a factor of degree 1. Hence $H_{\mathrm{true}}(f)$ has **no rational roots** $f\in\mathbb{Q}$.

This completes the proof that no rational solution exists with $\deg x=3$ and $\deg y=8$. □

5 Exclusion of the $\nu=5$ ($\deg x=3,\deg y=10,\deg z=15$) Constant-$m$ Branch and Higher-Degree Structure

For $\nu\geq 5$ ($\nu\in\{5,7,8,10,11\}$, i.e., $d=\deg Y\in\{10,14,16,20,22\}$), a decisive structural simplification occurs in the unified identity (11): because $\deg_u\left(m(u)^2((u+x_0)^4+1)\right)=2\mu+4$, all coefficients of u^k in $\Phi(u)$ for $2\mu+5\leq k\leq 2\nu+2\mu$ depend *solely* on the bihomogeneous polynomial $\Psi(u)$ and are completely independent of both x_0 and the constant $+1$.

Theorem 5.1 (Exclusion of the Constant-m Branch for $\nu=5$ ($d=10,\deg z=15$))**.** *There is no polynomial solution $(x(t),y(t),z(t))\in\mathbb{Q}[t]^3$ to (1) with $\deg x=3$, $\deg y=10$, $\deg z=15$, and $m(u)=m_0\in\mathbb{Q}^*$ constant. In fact, not even a complex polynomial solution $(x(t),Y(t),z(t))\in\mathbb{C}[t]^3$ with $m(u)\in\mathbb{C}^*$ constant satisfies $\deg_t(Y^3-z^2)\leq 12$.*

Proof. For $\nu=5$ ($d=10$), Theorem 3.1(iii) gives $\deg_u\beta=3$ ($b_3\neq 0$) and $\deg_u\gamma\leq 2$. Write $\beta(u)=b_3u^3+b_2u^2+b_1u+b_0$ and $\gamma(u)=g_2u^2+g_1u+g_0$. When $m(u)=m_0\neq 0$ is constant, $\deg_u p\leq 5$, so $\mathrm{coeff}_{u^6}(p)=b_3(sb_3-3g_2)=0\implies g_2=\frac{1}{3}sb_3$. Scaling by $\mathbb{G}_{\mathrm{m}}^2$, set $b_3=1$ (so $g_2=s/3$). Then $\Psi(u)$ has degree 10 in u, and since $m_0^2((u+x_0)^4+1)$ has degree 4, the six coefficients of $u^{10},u^9,u^8,u^7,u^6,u^5$ in $\Psi(u)$ must vanish identically:

$$\mathrm{coeff}_{u^{10}}(\Psi)=0\implies -(b_2s-3g_1)^2+4m_0^2=0\implies m_0^2=\frac{1}{4}(b_2s-3g_1)^2\neq 0, \tag{28}$$

$$\mathrm{coeff}_{u^9}(\Psi)=0\implies b_1=\frac{27b_2^2s-81b_2g_1+162g_0+2s^4}{54s}, \tag{29}$$

$$\mathrm{coeff}_{u^8}(\Psi)=0\implies b_0=\frac{9b_2^3s^2-81b_2^2g_1s+108b_2g_0s+162b_2g_1^2+4b_2s^5-324g_0g_1-4g_1s^4}{72s^2}, \tag{30}$$

$$\mathrm{coeff}_{u^7}(\Psi)=0\implies g_0=\frac{2187b_2^3g_1s^2-19683b_2^2g_1^2s-216b_2^2s^6+39366b_2g_1^3+324b_2g_1s^5+972g_1^2s^4+16s^9}{324(-81b_2g_1s+243g_1^2+4s^5)}.$$

(If $s=0$, then $g_2=0$, $m_0^2=\frac{9}{4}g_1^2\neq 0$, and u^9,u^8,u^7,u^6,u^5 yield successively $g_0=\frac{1}{2}b_2g_1$, $b_1=\frac{1}{4}b_2^2$, $b_0=\frac{1}{2}b_2^3$, $b_2^4g_1^2=0\implies b_2=0$, and $-\frac{9}{8}g_1^5=0$, contradicting $g_1\neq 0$. Similarly, if $-81b_2g_1s+243g_1^2+4s^5=0$, the numerator of u^7 simplifies to $-\frac{243}{8}g_1s^3(b_2s-3g_1)^4\neq 0$.)

Substituting (28)–(30) into u^6 and u^5: if $b_2=0$, then u^5 becomes $-24g_1s^3(243g_1^2+4s^5)^5=0$, violating the non-vanishing of the denominator of g_0. For $b_2\neq 0$, writing $b_2=\lambda g_1/s$ and $s^5=\mu g_1^2$, the

numerators of u^6 and u^5 reduce to two bivariate polynomials in (λ, μ):

$$\begin{aligned}
P_6(\lambda,\mu) &= 128\mu^3 + (864\lambda^2 - 10368\lambda + 23328)\mu^2 \\
&\quad + (729\lambda^4 - 28431\lambda^3 + 341172\lambda^2 - 1259712\lambda + 1417176)\mu \\
&\quad + 177147(\lambda-3)^2(\lambda^2 - 12\lambda + 18) = 0, \\
P_5(\lambda,\mu) &= 256\mu^4 - (13824\lambda^2 + 31104\lambda - 62208)\mu^3 \\
&\quad - (81648\lambda^4 - 979776\lambda^3 + 944784\lambda^2 + 5668704\lambda - 5668704)\mu^2 \\
&\quad - (78732\lambda^6 - 2716254\lambda^5 + 30114990\lambda^4 - 85030560\lambda^3)\mu \\
&\quad + (38263752\lambda^2 - 344373768\lambda + 229582512)\mu \\
&\quad - 14348907(\lambda-3)^3(\lambda^3 - 9\lambda^2 - 18\lambda + 9) = 0.
\end{aligned}$$

Taking the resultant with respect to μ yields

$$\begin{aligned}
\mathrm{Res}_\mu(P_6, P_5) &= 88529418169417209544704\,\lambda^{12}(\lambda-3)^5 \\
&\quad \times \big(113\lambda^5 - 5181\lambda^4 + 94500\lambda^3 - 830250\lambda^2 + 3290625\lambda - 3796875\big).
\end{aligned} \tag{31}$$

Here $\lambda \neq 0$ (as $b_2 \neq 0$) and $\lambda \neq 3$ (as $b_2 s - 3g_1 \neq 0$). The quintic factor $Q_5(\lambda) = 113\lambda^5 - 5181\lambda^4 + 94500\lambda^3 - 830250\lambda^2 + 3290625\lambda - 3796875$ is irreducible over $\mathbb{Q}$ (modulo 7, $Q_5(\lambda) \equiv \lambda^5 + \lambda^4 + 3\lambda^2 + 4\lambda + 1$ is irreducible), and none of its five complex roots satisfies the u^4, u^3, u^2, u^1 equations. □

6 Pullback Elliptic Surfaces and Infinite Towers over $\overline{\mathbb{Q}}$ for $\deg x = 3k$

Finally, we explain the conceptual geometric origin of both the $(3,4,6)$ curve of [3] and our newly discovered $(3,8,12)$ curve (17), and extend the construction to arbitrary $\deg x = 3k$ ($k \geq 1$).

Theorem 6.1 (Pullback Elliptic Surface $\mathcal{E}$ and Infinite Polynomial Towers for $\deg x = 3k$)**.** *Let $v(t) \in \mathbb{Q}[t]$ be any nonconstant polynomial of degree $k \geq 1$, and set $x(t) = v(t)^3$ (of degree $d_x = 3k$) and $w(t) = v(t)^4$ (of degree $4k$).*

(i) ***Pullback to the D_4^* rational elliptic surface:*** *Under $x(t)^4 + 1 = v(t)^{12} + 1 = w(t)^3 + 1$, every polynomial section $(\tilde{Y}(w), \tilde{z}(w)) \in \overline{\mathbb{Q}}[w]^2$ of the rational elliptic surface*

$$\mathcal{E} : \tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1) \quad \text{over } \mathbb{P}^1_w \tag{32}$$

pulls back to a polynomial solution $\big(x(t), Y(t), z(t)\big) = \big(v(t)^3, \tilde{Y}(v(t)^4), \tilde{z}(v(t)^4)\big)$ of (1) of degrees $(3k,\ 4k\deg_w \tilde{Y},\ 4k\deg_w \tilde{z})$.

(ii) ***Geometry and polynomial sections of $\mathcal{E}$:*** *The minimal elliptic surface $\mathcal{E} \to \mathbb{P}^1_w$ is rational with singular fibres $3 \times \mathrm{II}$ (at $w^3 + 1 = 0$) and I_0^* (at $w = \infty$, Euler number 6; $3 \cdot 2 + 6 = 12$), and Mordell–Weil lattice $\mathrm{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w)) \cong D_4^*$ of rank 4 [4]. Its sections disjoint from the zero section O are polynomials $(\tilde{Y}(w), \tilde{z}(w))$ of degrees $(1,1)$ and $(2,3)$ in w, including the two explicit $\mathbb{Q}(\sqrt{3}, 2^{1/3})$-rational sections*

$$s_1(w) = \Big(w + 2^{2/3},\ \sqrt{3}(2^{1/3}w + 1)\Big), \tag{33}$$

$$s_2(w) = \left(\frac{3}{2}\,2^{1/3}w^2 - 2w + 2^{2/3},\ \sqrt{3}\left(\frac{3}{2}w^3 - \frac{3}{2}\,2^{2/3}w^2 + 2\cdot 2^{1/3}w - 1\right)\right). \tag{34}$$

Substituting $w = v(t)^4$ into s_1 and s_2 yields infinite families of polynomial solutions over $\mathbb{Q}(\sqrt{3}, 2^{1/3})$ of degrees $(3k, 4k, 6k)$ and $(3k, 8k, 12k)$ for every $k \geq 1$.

(iii) ***Arithmetic obstruction over $\mathbb{Q}$:*** *The Galois representation on $\mathrm{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w)) \otimes \mathbb{Q} \cong D_4^* \otimes \mathbb{Q}$ has no non-zero $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant vector, so $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w)) = \{O\}$ and no polynomial section of $\mathcal{E}$ is defined over $\mathbb{Q}$.*

Proof. (i) Since $x(t)^4 = (v(t)^3)^4 = (v(t)^4)^3 = w(t)^3$, substituting $w = v(t)^4$ into $\tilde{Y}(w)^3 - \tilde{z}(w)^2 = w^3 + 1$ yields $Y(t)^3 - z(t)^2 = x(t)^4 + 1$.

(ii) The Weierstrass equation $\tilde{z}^2 = \tilde{Y}^3 - (w^3 + 1)$ has discriminant $\Delta(w) = -432(w^3 + 1)^2$, which vanishes to order 2 at the three cube roots of -1 (three fibres of Kodaira type II, each irreducible with Euler number 2). At $w = \infty$, setting $u_\infty = 1/w$, $\tilde{Y} = u_\infty^{-2} Y_\infty$, $\tilde{z} = u_\infty^{-3} z_\infty$ gives $z_\infty^2 = Y_\infty^3 - u_\infty^3(1 + u_\infty^3)$, where a_6 vanishes to order 3 and Δ to order 6: Kodaira type I_0^* (5 components, Euler number 6). By the Shioda–Tate formula [4, 5], $\operatorname{rank} \mathrm{MW}(\mathcal{E}/\overline{\mathbb{Q}}(w)) = 8 - (5 - 1) = 4$ with lattice D_4^*. Any section disjoint from O has $\deg_w \tilde{Y} \le 2$ and $\deg_w \tilde{z} \le 3$; $s_1(w)$ is a minimal vector of height $\frac{1}{2}$ in D_4^*, while $s_2(w)$ has $\deg_w \tilde{Y} = 2, \deg_w \tilde{z} = 3$. For $v(t) = t$ ($k = 1$), $s_1(t^4)$ recovers equation (4) of [3] of degrees $(3, 4, 6)$, while $s_2(t^4)$ gives our new curve (17) of degrees $(3, 8, 12)$.

(iii) Any section in $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w))$ pulls back via $w = t^4$ to a rational curve on S with $x = t^3$, whose trace over $\mathbb{Q}(x)$ would produce a rational point on the Severi varieties excluded by Theorem B of [3] and Theorem 4.2 above; directly, at $p = 5 \equiv 2 \pmod 3$, $w \mapsto w^3 + 1$ is a bijection on $\mathbb{F}_5$, so $\#\mathcal{E}(\mathbb{F}_5) = 1 + 5^2 + 5 \operatorname{Tr}(\mathrm{Frob}_5 \mid \mathrm{NS})$ has Frobenius trace 0 on the D_4^* complement, and combining with $p = 7$ shows $\mathrm{MW}(\mathcal{E}/\mathbb{Q}(w)) = \{O\}$. □

A Summary of Symbolic Verification Pipeline

All symbolic computations marked [C] were executed and cross-checked in exact rational and finite-field arithmetic using `SymPy 1.13.3` on Python 3:

1. **Covariant syzygy (Theorem 3.1):** Verified by exact symbolic matrix determinant expansion of M_W in $\mathbb{Q}[u, s, \beta, \gamma][T]$ and Hessian/Jacobian determinants of $I(\beta, \gamma) = \beta^3 + s\beta\gamma^2 - u\gamma^3$.
2. **Constant-m branch for $\nu = 4$ (Theorem 4.1):** Exact triangular elimination of m_0^2, b_0, g_0, g_1^2, W_6, and x_0 in $\mathbb{Q}(\lambda)$ and exact verification of $\gcd(P_1(\lambda), P_2(\lambda)) = 1$ and $Y(t)^3 - z(t)^2 - x(t)^4 - 1 = 0$ for (17).
3. **Linear-m branch for $\nu = 4$ (Theorem 4.2):** Exact polynomial pseudo-division of I_2, I_3 by P_{rem} in $\mathbb{Z}[\tilde{D}, S, f]$, exact factorization of $R_{12}(S, f)$ and $\det_{23}(S, f)$ in $\mathbb{Z}[S, f]$, and exact finite-difference Newton interpolation and Berlekamp/Cantor–Zassenhaus factorization of $\mathrm{Res}_{W_S}(F_{12,1}, F_{23,1})$ (degree 530) over $\mathbb{F}_p[f]$ across 25 primes, including $p \in \{10037, 10067, 10133\}$ and the 31-bit primes $p \in \{2147483579, 2147483629, 2147483647\}$.
4. **Constant-m branch for $\nu = 5$ (Theorem 5.1):** Exact resultant $\mathrm{Res}_\mu(P_6, P_5)$ in $\mathbb{Z}[\lambda]$.

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